

Yu-Yun Chang

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MODEL RISK · QUANTITATIVE VALIDATION · MACRO-FINANCIAL FORECASTING

PROFILE

Economics PhD candidate with an M.S. in Financial Engineering, specializing in macro-financial forecasting, time-series econometrics, and machine learning. Seven years of quantitative research under out-of-sample discipline, with documented data lineage, pre-specified test plans, and formal change control. CFA Level I. Seeking a model risk or quantitative validation role.

EDUCATION

Ph.D. in Economics, The Graduate Center, City University of New York, NY Expected June 2027
Dissertation: *Measuring the Exchange-Rate Disconnect: Sample, Timing, and Aggregation*
Advisor: Prof. Merih Uctum

M.S. in Financial Engineering, NYU Tandon School of Engineering, New York, NY 2018
Coursework: Computational Finance · Derivatives Pricing · Risk management

M.A. in International Economics, National Chung Cheng University, Taiwan 2015

B.B.A. in Economics, National Central University, Taiwan 2011

TECHNICAL SKILLS

Model Validation & Governance — Conceptual soundness assessment; implementation replication; champion–challenger benchmarking; out-of-sample and rolling-window backtesting; multiple-testing correction; detectability and power analysis; sensitivity, subsample, and stress analysis; data lineage and provenance audit; pre-specified test registration; frozen change-control protocols; graded findings and limitations documentation

Econometrics & Time Series — Integration order testing and break-robust methods; cointegration (Johansen, ARDL-ECM); VAR and VECM; threshold and regime-switching models; time-varying parameter models; state-space methods; ARIMA; GARCH-family volatility modeling; panel estimation and cross-sectional dependence testing; bootstrapped inference; difference-in-differences

Machine Learning — XGBoost; random forest; LASSO and shrinkage ridge; neural networks; ensemble methods; SHAP attribution; walk-forward cross-validation

Programming & Data Engineering — Python (pandas, scikit-learn, statsmodels, XGBoost, arch); SQL and PostgreSQL; R; Stata; Git; LaTeX; Bloomberg; FRED/ALFRED real-time vintage data; OECD

Financial Markets & Risk — Foreign exchange; monetary policy and macro-financial transmission; global risk factors (volatility, credit spreads, term premia); Value-at-Risk and Expected Shortfall; derivatives

QUANTITATIVE RESEARCH & MODEL VALIDATION PRACTICE

Beyond Data Release Timing: Information Alignment and the Measurement of Exchange-Rate Predictability (*job market paper*)

- Built a pseudo-real-time forecasting framework across 14 currency pairs and four global risk factors, comparing OLS, LASSO, random forest, and XGBoost against random-walk and historical-mean benchmarks under out-of-sample evaluation.
- Identified a calendar-timing defect in target construction through a data-provenance audit; retired all affected results as non-citable, rebuilt the pipeline under corrected alignment rules, and preserved the original outputs as a forensic record.
- Enforced information alignment structurally through a machine-readable timing dictionary read at execution; pre-registered three confirmatory hypotheses and froze the architecture before opening clean results, reporting all three null outcomes as specified.

- Designed a timing counterfactual isolating clean versus contemporaneous predictor alignment on identical realized returns, converting a data-handling correction into a measurable quantity.
- Replicated five pre-frozen canonical predictor families (UIP, PPP, Taylor rule, monetary, NFA) across 62 currency-family cells under both alignment regimes with Holm multiplicity correction; no family showed a significant timing effect, and a built-in mechanical placebo confirmed the design.
- Traced the result to information renewal rather than publication delay: the largest alignment effect arises from a predictor with no publication lag, while highly persistent macro predictors show small effects, linked by an identity relating standardized one-period updates to persistence.

Exchange Rate Adjustment and Reconnection to Fundamentals (*dissertation chapter*)

- Froze a Stage-0 routing and change-control protocol before any estimation, deferring model-family selection (Johansen/VECM, ARDL-ECM, or no-anchor) to a data audit rather than assuming a specification.
- Built a frozen economic-time panel of 14 USD pairs on a four-variable baseline (log exchange rate, interest differential, relative prices, relative industrial production) with currency-specific effective samples and documented source exceptions.
- Ran an integration, break, and cross-sectional dependence audit: all 14 exchange rates classified I(1), fundamentals heterogeneously; no currency obtained primary support for a stable long-run relationship, and the short-run stage was likewise null.
- Closed a detectability extension as an analytical benchmark after terminating a disproportionate simulation design: at a literature-calibrated adjustment speed implying a multi-year half-life, no currency reaches the conventional inference threshold, bounding what the available samples can resolve.

Unit of Analysis and Pooled Identification (*dissertation chapter*)

- Tests whether pooling across currencies recovers structural and predictive content that single-currency estimation cannot detect, reusing the two frozen panels with no new data construction.

APPLIED PROJECTS

Independent Validation of a GARCH Volatility Model

[GitHub] · [Date]

Independent validation of a GARCH(1,1) model used for Value-at-Risk: conceptual soundness and parameter stability assessment, replication from first principles against a reference implementation, benchmarking against EWMA and asymmetric-GARCH challengers, and VaR backtesting under Kupiec, Christoffersen, and Basel traffic-light criteria. Delivered graded findings with remediation recommendations.

Point-in-Time Financial Data Governance Pipeline

[GitHub] · [Date]

FRED/ALFRED vintage data with release-date-aware joins, schema validation, publication-lag profiling, and automated data-quality reporting. Quantified the backtest performance gap between final-revised and point-in-time data regimes. Deployed on AWS (S3, RDS, Lambda, EventBridge) under least-privilege IAM.

Independent Validation of a Retrieval-Augmented LLM System

[GitHub] · [Date]

Validation framework for a RAG system over regulatory guidance documents, built on a hand-verified evaluation set spanning in-scope, out-of-scope, and adversarial cases. Measured retrieval recall, answer faithfulness, and hallucination rate; ran prompt-sensitivity and non-determinism stability tests; produced a failure taxonomy, a model card, and an SR 11-7-structured validation report. Built on Amazon Bedrock with S3 Vectors.

Model Monitoring and Inventory Control Center

[GitHub] · [Date]

Integration layer tracking models through a tiered inventory with materiality-based validation cycles. Monitors input drift (PSI, Kolmogorov-Smirnov), tracks performance decay against control limits, segments performance by volatility regime, and drives three-tier alerting with a remediation log. Automated reporting via AWS Lambda, CloudWatch, and SNS.

PUBLICATION & PRESENTATION

Hung, J. and Y.-Y. Chang. “Why Has the United States Continued to Receive Net Investment Income as a Debtor Country?” *The World Economy*, 42(3), March 2019.

Western Economic Association International Annual Conference, San Diego, 2017.

EARLIER INDUSTRY EXPERIENCE

Financial Analyst — Epic Commercial Realty, New York, NY Feb 2018 – Feb 2019

- Performed valuation, financial statement analysis, and due diligence for commercial real estate acquisitions; prepared investment memoranda covering expected returns, market conditions, and transaction risk.

Financial Analyst — Shan-Jhu Construction, Taipei, Taiwan Jul 2012 – Jul 2015

- Prepared investment reports and portfolio analyses for senior portfolio managers. Monitored macroeconomic indicators and market developments to identify exposures and opportunities.

TEACHING

Graduate Teaching Fellow — Hunter College, CUNY 2020–2024

Mathematics for Economics, Nonlinear Econometric Analysis, Microeconomic Theory, Macroeconomic Theory, and International Trade and Finance.

Taught theoretical and empirical methods and advised on research design, causal inference, and written presentation of quantitative results.

CERTIFICATIONS, AWARDS & ADDITIONAL

Certifications: CFA Level I · AWS Generative AI and AI Agents with Amazon Bedrock (Professional) · IBM Building AI Agents and Agentic Workflows

Awards: Graduate Center Fellowship, CUNY (2019) · Graduate Scholarship, NYU Tandon (2016) · Graduate Scholarship, National Chung Cheng University (2014)

Languages: Mandarin Chinese (native) · English (fluent)